Reddy Roshni

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.17
Sri Chaitanya Jr Kalasala-Parasara Bhavan	Class XII (TSBIE)	2023	98.90%
Carmel Convent High School	Class X (CBSE)	2021	90.00%

WORK EXPERIENCE

Samsung R&D Institute India, Noida, Health SDK Team | Summer Intern

- Reverse-engineered a third-party SDK's integration architecture end to end, mapping how in-app messaging, push notifications, event tracking, and Content Cards propagate into downstream analytics pipelines.
- Built asynchronous data-retrieval components using Kotlin Coroutines, reasoning explicitly about lifecycle-aware execution to avoid race conditions across async flows.
- Designed a standalone demo application to isolate and verify BroadcastReceiver behavior across Android lifecycle transitions, converting ambiguous async behavior into a reproducible test case.

PROJECTS

Online Code Judge | Node.js, Express.js, MongoDB, C++, REST APIs

- Built a backend online judge with REST APIs and Controller-Service-Repository architecture, managing problems, testcases, and submissions in MongoDB.
- Implemented a C++ execution pipeline using Node.js child_process.exec() for compilation and child_process.spawn() for binary execution, passing testcase input through stdin and capturing stdout/stderr.
- Implemented a 2-second execution timeout using setTimeout() and child.kill(); handled compilation failures and non-zero exit codes, with temporary source/executable cleanup using Node.js fs APIs.
- Automated submission evaluation across Accepted, Wrong Answer, Compilation Error, Runtime Error, and Time Limit Exceeded, with submission status persisted in MongoDB.

String Matching Algorithms | C++, C, Data Structures & Algorithms | Team project

- Implemented classic pattern-matching algorithms including Rabin-Karp hashing, Finite Automata, Knuth-Morris-Pratt, and Naive Suffix Trees for efficient pattern search in text.
- Implemented Arithmetic Encoding as a subroutine for a PPM data compression model prototype for strings of small length, and developed LZ77 data compression using Suffix Trees to detect repeated substrings.

Statistical Analysis of Sleeping Habits of IITH Students | Team Project

- Ran a Google Forms survey of 342 IIT Hyderabad students on sleep habits and academic impact; built the Python analysis and visualization pipeline.
- Found weekday sleep averaged 6.64 h vs. 7.90 h on weekends, and a one-proportion z-test showed 64.9% had missed academic work due to tiredness ($z = 5.52$, $p < 0.001$); fitted a Gamma distribution (MoM and MLE) and computed 95% confidence intervals.

SKILLS

- **Languages & Scripting:** C++, SQL, C, JavaScript (Node.js)
- **Infrastructure & Systems:** Process Scheduling, Inter-Process Communication (IPC), OS Resource Management, Timeout Enforcement
- **Tools & Containers:** Git, GitHub, Docker (Containers), REST APIs, Postman
- **Databases & Analytics:** MySQL, MongoDB, Python (NumPy, Pandas, SciPy)
- **Problem Solving:** Solved 600+ DSA problems across LeetCode and GeeksforGeeks (Striver's A2Z DSA Sheet)

POSITIONS OF RESPONSIBILITY

Sunshine Buddy, Sunshine - The Counselling Cell of IIT Hyderabad (2024-2025)

Supported student engagement and peer interaction as part of Sunshine, contributing to student-support initiatives and maintaining regular communication with peers.

EXTRACURRICULAR

Core Member, Chess Team, NSO, IIT Hyderabad (2023-Present)

Medalist in Badminton, representing Mathematics and Computing at DIESTA 2024, IIT Hyderabad

RELEVANT COURSEWORK

Data Structures & Algorithms, Database Management Systems, Operating systems, Theory of Computation, Artificial Intelligence, Probability Theory, Linear Algebra, Combinatorics & Graph Theory, Number Theory, Applied Statistics, Convex Optimization